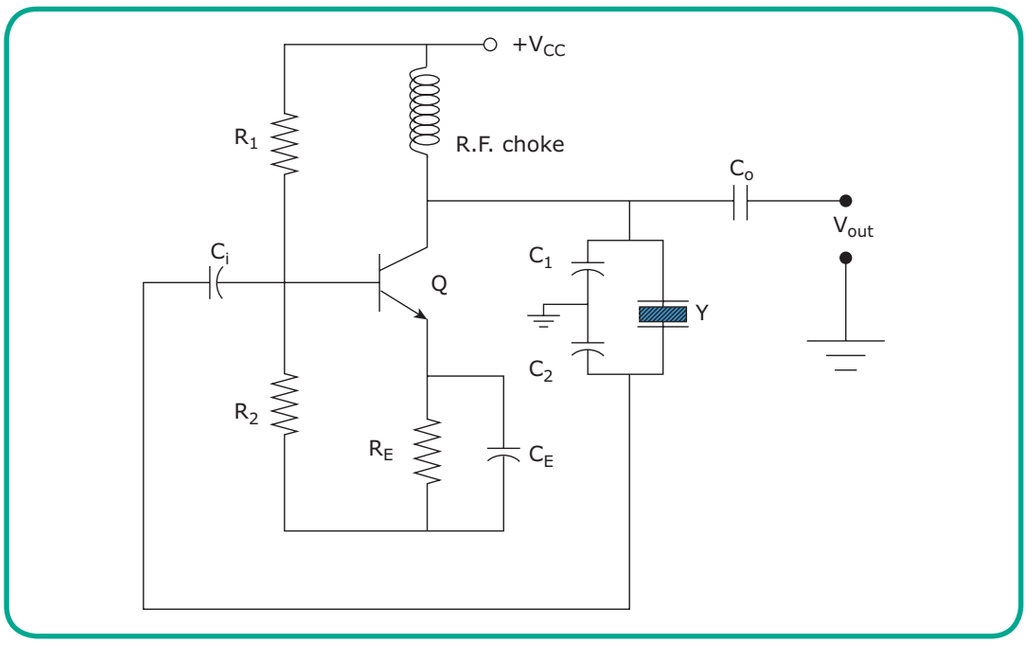


Figure 7.8 Crystal Oscillator

the converse effect, an electric field results in mechanical strain. The advantage of the crystal is its very high Q as a resonant circuit, which results in good frequency stability for the oscillator.

Circuit Description: In the circuit shown in Figure 7.8, the resistors R_1 and R_2 form the voltage divider network, while the emitter resistor R_E stabilizes the circuit. Further, C_E acts as an AC by pass capacitor, while the coupling capacitor C_i is used to block DC signal propagation between collector and base terminals. The RF coil in the circuit which offers dual advantages, as it provides even the DC bias and frees the circuit output from being affected by the AC signal on the power lines.

Working Principle: When the supply is switched 'ON', the capacitor C_1 is charged. It fully charged, then starts discharging through the crystal and produces oscillation. The frequency of the oscillation depends upon the values of C_1, C_2 and the RLC values of the crystal. If the frequency of the oscillation is equal to its crystal resonant frequency, the circuit produces more stable oscillations. The crystal frequency is dependent on temperature.

Advantages

1. High 'Q' factor
2. Excellent frequency stability
3. Simple circuit

Disadvantages

1. Not to be used as tuned oscillators
2. Crystal is fragile type, hence it is used only in low power circuits

7.5.4. Application of crystal oscillator

Crystal oscillators are used in the microprocessor and microcontroller for providing the clock signals. It generates clock pulses required for the synchronization of all the internal operations. The use of crystal oscillator in military and aerospace is to establish an efficient communication system for navigation purpose in the guidance systems.

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